Comparative Effectiveness of Diabetes Treatments Using Synthetic Real-World Data

Real-World Evidence Report — Synthetic RWE Diabetes Treatment Study

Comparative Effectiveness: Metformin vs Other Diabetes Medications

Report generated: 2026-07-21
Final analytic cohort: N = 69

Abstract. This real-world evidence report describes a comparative effectiveness study of metformin versus other diabetes medications in 69 adults with type 2 diabetes using Synthea synthetic EHR data. Propensity score matching, inverse probability weighting, and regression adjustment were applied to estimate treatment effects on a composite outcome of diabetes complications or hospitalization. Unadjusted event rates were higher in the other-medication group; adjusted analyses attenuated but did not eliminate differences. Causal interpretation requires no unmeasured confounding and adequate overlap.

1. Introduction

Research question: What is the causal effect of diabetes pharmacotherapy (metformin vs other diabetes medications) on the composite outcome of diabetes-related complications or hospitalization?

Type 2 diabetes mellitus requires long-term pharmacotherapy. Metformin is typically first-line therapy, but patients with advanced disease or contraindications may initiate other agents (e.g., insulin). Observational comparative effectiveness research can inform treatment decisions when randomized trials are unavailable, provided rigorous causal methods address confounding by indication.

2. Methods

2.1 Study design and data source

- **Design:** Retrospective observational cohort study with propensity score and regression-based causal inference.
- **Data source:** Synthea synthetic electronic health record (EHR) data (conditions, medications, encounters).
- **Population:** Adults with type 2 diabetes initiating diabetes pharmacotherapy after diagnosis; final cohort N = 69 (49 metformin, 20 other medication at index).
- **Treatment comparison:** Metformin users vs Other diabetes medication users (assigned at first post-diagnosis diabetes medication).

- **Follow-up:** Up to 730 days from index treatment date (administrative censoring at last encounter or death). Mean follow-up ≈ 8.3 years.

- **Primary outcome:** Composite of first post-index diabetes complication, cardiovascular event, or hospitalization/emergency visit

2.2 Statistical analysis

Baseline characteristics were compared using Table 1 with standardized mean differences (SMD). Outcomes were summarized as event counts, percentages, and rates per 100 patient-years. Kaplan-Meier and log-rank tests assessed time to primary composite event. Causal effects were estimated using 1:1 propensity score matching, stabilized IPTW (ATE and ATT), and multivariable logistic regression, supplemented by doubly robust AIPW (Step 7).

3. Results

3.1 Study population and cohort flow

The analytic cohort comprised 69 patients selected through sequential eligibility criteria applied to the Synthea patient population (Figure 1).

Figure 1. Study Cohort Flow

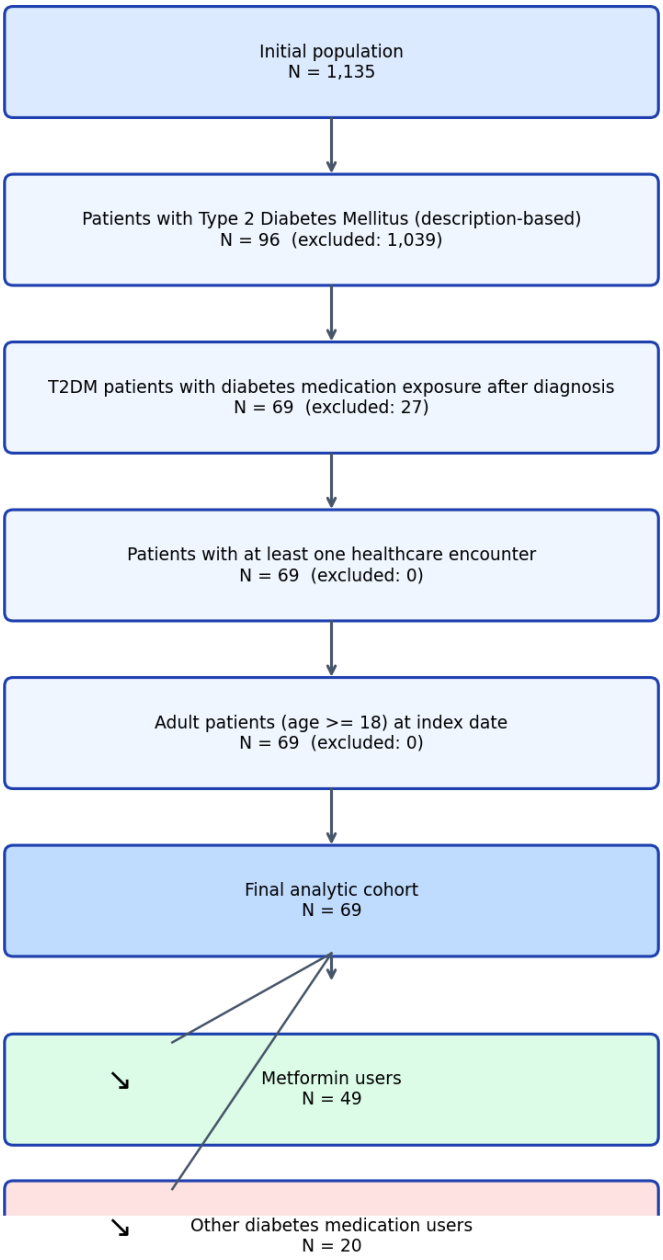


Figure 1. Patient selection flow from initial population to final analytic cohort and treatment groups.

3.2 Baseline characteristics (Table 1)

Baseline characteristics differed between treatment groups before adjustment. 3 of 17 evaluated variables achieved |SMD| < 0.1 (18%), indicating incomplete balance at baseline. Older age, higher comorbidity burden, and greater diabetic nephropathy prevalence in the other-medication group suggest confounding by indication.

Category	Variable	Metformin users	Other diabetes medication users	Overall (N=69)	SMD	Balanced (|SMD|<0.1)	P-value	Sig.
Demographics	N	49	20	69	0.000	True	NaN	NaN
Demographics	Age, years — mean (SD)	54.8 (11.8)	69.4 (16.1)	59.1 (14.7)	-1.036	False	0.0003	***
Demographics	Age, years — median (IQR)	51.2 (47.6–63.2)	66.1 (56.1–81.5)	55.6 (49.4–66.8)	NaN	NaN	NaN	NaN
Demographics	Age, years — min / max	36.3 / 95.9	44.6 / 99.8	36.3 / 99.8	NaN	NaN	NaN	NaN
Demographics	Gender: F	24 (49.0%)	13 (65.0%)	37 (53.6%)	-0.322	False	0.3448	NS
Demographics	Gender: M	25 (51.0%)	7 (35.0%)	32 (46.4%)	0.322	False	NaN	NaN
Demographics	Race: asian	1 (2.0%)	0 (0.0%)	1 (1.4%)	0.202	False	0.4631	NS
Demographics	Race: black	2 (4.1%)	3 (15.0%)	5 (7.2%)	-0.370	False	NaN	NaN
Demographics	Race: hawaiian	1 (2.0%)	0 (0.0%)	1 (1.4%)	0.202	False	NaN	NaN
Demographics	Race: native	1 (2.0%)	0 (0.0%)	1 (1.4%)	0.202	False	NaN	NaN

Demographics	Race: white	44 (89.8%)	17 (85.0%)	61 (88.4%)	0.142	False	NaN	NaN
Clinical characteristics	Hypertension	27 (55.1%)	12 (60.0%)	39 (56.5%)	-0.097	True	0.9166	NS
Clinical characteristics	Hyperlipidemia	8 (16.3%)	4 (20.0%)	12 (17.4%)	-0.094	True	0.7343	NS
Clinical characteristics	Obesity	49 (100.0%)	14 (70.0%)	63 (91.3%)	0.902	False	0.0003	***
Clinical characteristics	Cardiovascular disease	11 (22.4%)	10 (50.0%)	21 (30.4%)	-0.587	False	0.0490	*
Clinical characteristics	Chronic kidney disease	7 (14.3%)	13 (65.0%)	20 (29.0%)	-1.188	False	0.0001	***
Clinical characteristics	Diabetic nephropathy	7 (14.3%)	16 (80.0%)	23 (33.3%)	-1.716	False	0.0000	***
Clinical characteristics	Diabetic neuropathy	5 (10.2%)	5 (25.0%)	10 (14.5%)	-0.388	False	0.1399	NS
Clinical characteristics	Diabetic retinopathy	1 (2.0%)	1 (5.0%)	2 (2.9%)	-0.158	False	0.4987	NS

3.3 Treatment exposure

Treatment exposure was defined at the first diabetes medication after T2DM diagnosis (index date). During follow-up, 26.1% of patients switched medication class and 26.1% used multiple diabetes drug classes when switchers are included.

- Exposure inferred from medication orders, not confirmed adherence.

- Treatment switching during follow-up may dilute intention-to-treat effects.

- Index-date assignment may misclassify patients who initiated non-metformin therapy before metformin in the same window.

3.4 Clinical outcomes (Table 2)

Unadjusted outcome rates for the primary composite were Metformin users: 37/49 events (75.5%; 8.9 per 100 patient-years); Other diabetes medication users: 18/20 events (90.0%; 11.6 per 100 patient-years).

Outcome	Treatment group	N patients	N events	Event rate (%)	Person-y	Events per 100 PY	Odds ratio	95% CI	P-value	Mean follow-up (years)	Median follow-up (years)	Median event-free days	Log-rank p-value
Cardiovascular Event	Metformin users	49	13.0	26.5	414.7	3.13	—	—	—	—	—	—	—
Cardiovascular Event	Other diabetes medication users	20	5.0	25.0	155.9	3.21	—	—	—	—	—	—	—
Cardiovascular Event	Overall	69	18.0	26.1	570.6	3.15	—	—	—	—	—	—	—
Diabetes Complication	Metformin users	49	16.0	32.7	414.7	3.86	—	—	—	—	—	—	—
Diabetes Complication	Other diabetes medication users	20	10.0	50.0	155.9	6.42	—	—	—	—	—	—	—
Diabetes Complication	Overall	69	26.0	37.7	570.6	4.56	—	—	—	—	—	—	—

Hospitalization Event	Metformin users	49	33.0	67.3	414.7	7.96	—	—	—	—	—	—	—
Hospitalization Event	Other diabetes medication users	20	17.0	85.0	155.9	10.91	—	—	—	—	—	—	—
Hospitalization Event	Overall	69	50.0	72.5	570.6	8.76	—	—	—	—	—	—	—
Primary Composite Event	Metformin users	49	37.0	75.5	414.7	8.92	—	—	—	—	—	—	—
Primary Composite Event	Other diabetes medication users	20	18.0	90.0	155.9	11.55	—	—	—	—	—	—	—
Primary Composite Event	Overall	69	55.0	79.7	570.6	9.64	—	—	—	—	—	—	—
Primary Composite Event (unadjusted OR)	Other vs Metformin	69	55.0	—	—	—	2.92	0.59–14.45	0.1893	—	—	—	—
Diabetes Complications (unadjusted OR)	Other vs Metformin	69	26.0	—	—	—	2.06	0.71–5.96	0.1810	—	—	—	—
Cardiovascular Event (unadjusted OR)	Other vs Metformin	69	18.0	—	—	—	0.92	0.28–3.05	0.8955	—	—	—	—
Hospitalization Event (unadjusted OR)	Other vs Metformin	69	50.0	—	—	—	2.75	0.70–10.76	0.1467	—	—	—	—
Follow-up duration (years)	Metformin users	49	—	—	—	—	—	—	—	8.46	10.00	—	—
Follow-up duration (years)	Other diabetes medication users	20	—	—	—	—	—	—	—	7.79	10.02	—	—

Kaplan-Meier (primary composite)	Metformin users	49	37.0	—	—	—	—	—	—	—	—	945.0	0.0101
Kaplan-Meier (primary composite)	Other diabetes medication users	20	18.0	—	—	—	—	—	—	—	—	459.0	0.0101

Kaplan-Meier analysis (primary composite): Kaplan-Meier curves for the primary composite differed by treatment group (log-rank p = 0.0101). Median event-free survival was longer in the metformin group.

3.5 Causal treatment effects (Table 3)

Table 3 summarizes causal effect estimates from propensity score matching (Step 6), IPTW and regression adjustment (Step 7). Effect measures include odds ratios (Other vs Metformin), risk differences (Metformin – Other), ATE, and ATT.

Analysis method	Effect measure	Estimate	95% CI	P-value	ATE (risk diff.)	ATT (risk diff.)	Odds ratio (Other vs Metformin)	N	Matched pairs	OR 95% CI	Weighted risk Metformin	Weighted risk Other
Propensity score matching (1:1 NN)	Odds ratio (Other vs Metformin)	0.2940	0.01–8.366	0.4737	0.2940	—	0.294	16	8.0	—	—	—
Regression adjustment	Odds ratio (Other vs Metformin)	1.1950	0.101–14.153	0.8878	—	—	1.195	69	—	—	—	—

Inverse probability of treatment weighting (IPTW)	ATE — risk difference (Metformin – Other)	0.1058	—	0.3384	0.1058	—	0.576	69	—	0.186–1.781	0.7896	0.6837
IPTW — average treatment effect on the treated (ATT)	ATT — risk difference (Metformin – Other)	0.1407	—	0.2090	—	0.1407	0.517	69	—	0.184–1.447	—	—
Doubly robust (AIPW)	Average treatment effect (risk difference)	0.1955	—	0.2399	—	—	0.236	69	—	0.021–2.626	—	—

Statistical interpretation. After adjustment for measured baseline characteristics, treatment group differences in the primary composite outcome were attenuated relative to unadjusted analyses. Multivariable logistic regression yielded an adjusted odds ratio of 1.195 (Other vs Metformin), with wide confidence intervals reflecting the modest sample size. Propensity score matching on 8 pairs produced unstable effect estimates due to small matched sample size and outcome separation. The estimated treatment effect represents the association between treatment exposure and clinical outcomes under causal assumptions (exchangeability, positivity, consistency), not a definitive causal claim given synthetic data and residual confounding.

4. Discussion

This synthetic RWE study demonstrates a full pharmaceutical observational analysis pipeline from data quality through evidence generation. Unadjusted analyses suggested higher composite event rates among patients initiating non-metformin therapy, consistent with confounding by indication (older, more comorbid patients). After multivariable and propensity-based adjustment, effect estimates moved toward the null with wide uncertainty intervals.

5. Limitations

- Synthetic data source (Synthea): results may not reflect real-world treatment patterns, coding practices, or clinical heterogeneity in live EHR/claims databases.

- Residual confounding: unmeasured disease severity (e.g., HbA1c, insulin resistance) may persist despite multivariable and propensity score adjustment.

- Missing variables: laboratory values, medication adherence, and lifestyle factors were not fully captured in the synthetic dataset.

- Medication adherence uncertainty: pharmacy fill data proxy exposure but do not confirm ingestion.

- Small sample size (N=69) limits statistical power, subgroup analyses, and precision of effect estimates.

- Generalizability: findings from a single synthetic cohort cannot be extrapolated to specific healthcare systems or patient populations without external validation.

6. Conclusion

Results estimate the causal effect under assumptions of no unmeasured confounding, correct model specification, and adequate overlap of treatment groups. While directionally consistent with a protective association for metformin, the small synthetic cohort precludes definitive clinical conclusions. External validation in real-world data is required before informing treatment decisions.